

POWER BI PROJECT

Understanding &
setting metrics and KPIs

Data visualization &
telling stories with data

HR Workforce & Attrition Analysis Dashboard

Strategic Workforce Intelligence for Data-Driven HR Decisions

 Interactive Analytics  Workforce Insights  Actionable Intelligence

Reading &
interpreting reports

Understanding the
relationship between
people data & the business

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Project Overview

Transforming workforce data into actionable business intelligence



Project Objective

Analyze employee workforce trends, attrition behavior, and demographic insights using an interactive Power BI dashboard to provide actionable HR business intelligence for better workforce management.

- ✓ Enable data-driven HR decisions
- ✓ Monitor workforce health metrics
- ✓ Identify retention opportunities



Dataset Domain

Human Resources / HR Analytics domain focusing on workforce and employee records including demographics, department details, hiring data, performance scores, and employment status.



Three Interactive Dashboards

1

Executive Overview Dashboard

High-level HR performance metrics and organizational workforce health assessment for leadership decision-making.

2

Demographics & Reasons Analysis

Deep dive into employee demographics, termination reasons, and recruitment source effectiveness analysis.

3

Employee Details Dashboard

Individual employee lookup with drill-down capabilities for detailed HR review and analysis.

Problem Statement

Understanding the critical challenges in workforce management



Organizational Challenges

Organizations frequently struggle to gain clear visibility into their workforce dynamics. **Without structured analytics, HR teams lack the insights needed to make proactive, data-driven decisions** about talent retention and workforce planning.

The Impact:

- Reactive rather than proactive HR management
- High recruitment and training costs due to turnover
- Loss of institutional knowledge and expertise
- Decreased team morale and productivity



Project Solution

This project addresses these challenges by **building structured Power BI dashboards** that transform raw HR data into visual, actionable insights—supporting strategic HR decision-making and enabling proactive workforce management.

? Key Questions to Answer

1

Why are employees leaving?

Identify primary termination reasons and patterns

2

Which departments face high attrition?

Pinpoint departments with retention challenges

3

Which recruitment channels are effective?

Evaluate recruitment source performance and ROI

4

Does performance affect attrition?

Analyze correlation between performance and turnover

5

How do demographics influence stability?

Understand demographic factors in workforce retention

Dataset & Technology Stack

Data sources and analytical tools powering the insights



HR Employee Dataset



Employee Demographics

Age, gender, marital status, race, date of birth



Department & Position Details

Department, position title, manager name



Hiring & Termination Data

Date of hire, termination date, employment status



Recruitment Sources

Indeed, LinkedIn, Google Search, referrals, etc.



Performance Scores

Fully Meets, Exceeds, Needs Improvement, PIP



Compensation Data

Salary information and pay structure



Data Quality: Dataset was already clean and properly structured—no major data transformation required.



Tools & Technologies



Microsoft Power BI Desktop

Primary visualization and dashboard development platform with interactive reporting capabilities.



DAX (Data Analysis Expressions)

Custom calculations and measures for Total Employees, Active Employees, Terminated Employees, and Attrition Rate.



Interactive Visualizations

Slicers, filters, drill-down capabilities, and dynamic page navigation for user-driven exploration.



Power BI Service

Cloud-based publishing and sharing platform enabling real-time access and collaboration.

04 Methodology & Implementation Steps

Systematic workflow from data import to dashboard deployment

1 Data Import Step 01

Imported HR dataset into Power BI from the clean, structured source file. Established data connections and verified table relationships for seamless analysis.

✓ Direct data connection established

2 Data Quality Check Step 02

Verified dataset quality by checking for missing values, data consistency, and proper formatting. Confirmed data integrity across all employee records.

✓ No major transformation needed

3 DAX Measures Step 03

Created DAX measures for key workforce metrics including Total Employees, Active Employees, Terminated Employees, and Attrition Rate calculations.

✓ 4 core measures defined

4 Dashboard Design Step 04

Designed three interactive dashboards with appropriate visualizations, charts, and KPI cards tailored to each dashboard's analytical purpose.

✓ Visual hierarchy established

5 Interactive Features Step 05

Added slicers for dynamic filtering by department, employee name, and other dimensions. Implemented page navigation for seamless user experience.

✓ User interactivity enabled

6 Cloud Deployment Step 06

Published dashboard to Power BI Service for cloud-based access, enabling real-time collaboration and sharing across the HR organization.

✓ Live dashboard deployed

Key DAX Measures Created



Total Employees
COUNT(EmployeeID)

Active Employees
Status = "Active"

Terminated
Status ≠ "Active"

Attrition Rate
Terminated/Total

Executive Overview Dashboard

High-level HR performance metrics for leadership decision-making

Key Performance Indicators

311

Total Employees

207

Active Employees

104

Terminated

33%

Attrition Rate

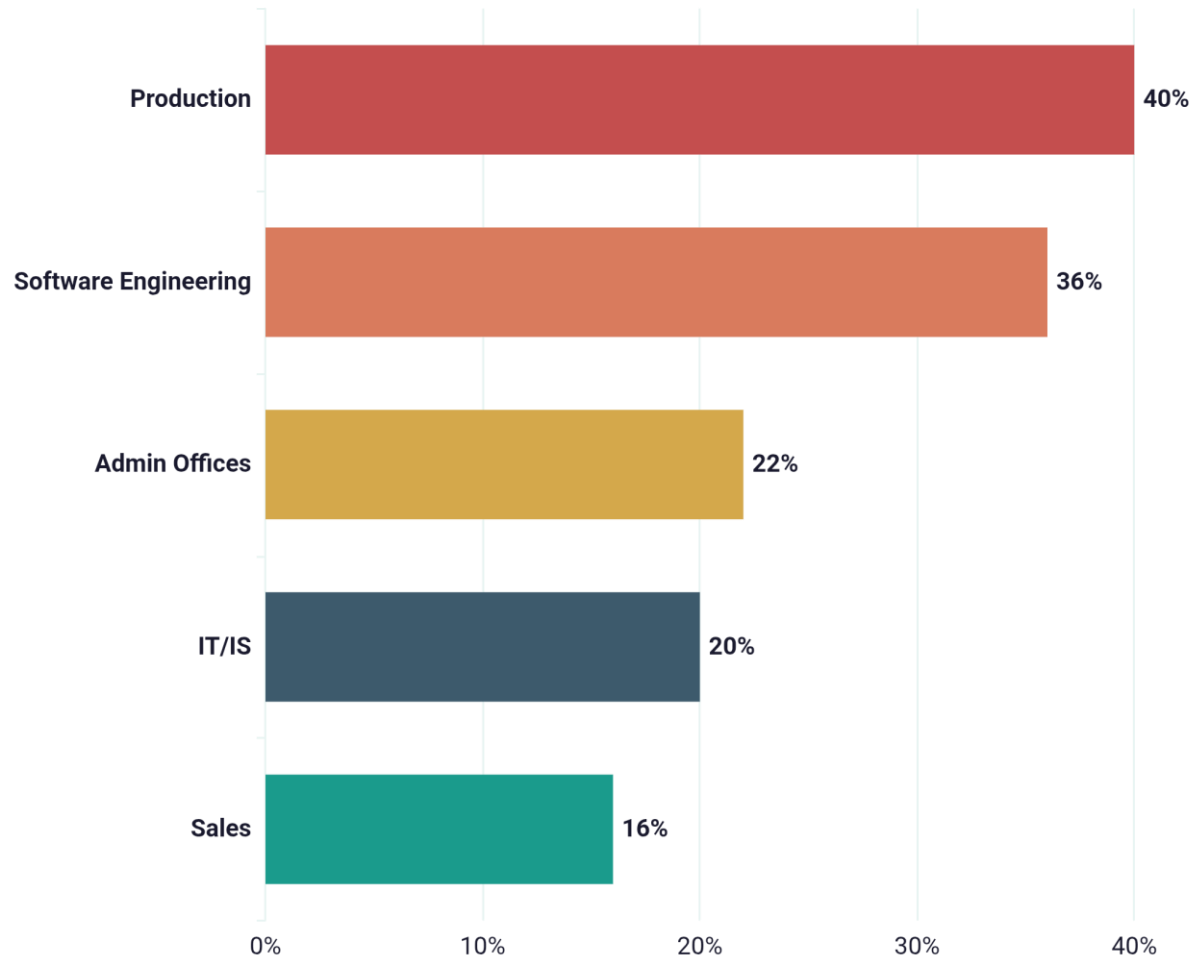
Dashboard Purpose

This dashboard provides **leadership with a quick, comprehensive assessment of organizational workforce health** through high-level metrics and visual trends.

Key Benefits:

- ✓ Real-time workforce visibility
- ✓ Department-level attrition tracking
- ✓ Performance correlation analysis
- ✓ Recruitment source evaluation

Attrition Rate by Department

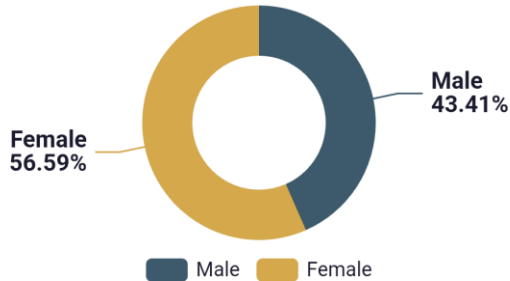


06

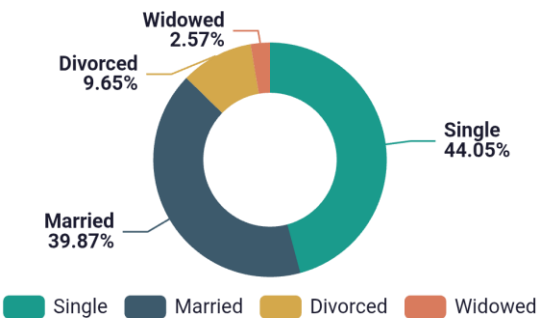
Demographics & Reasons Analysis

Understanding who is leaving and why through demographic breakdown

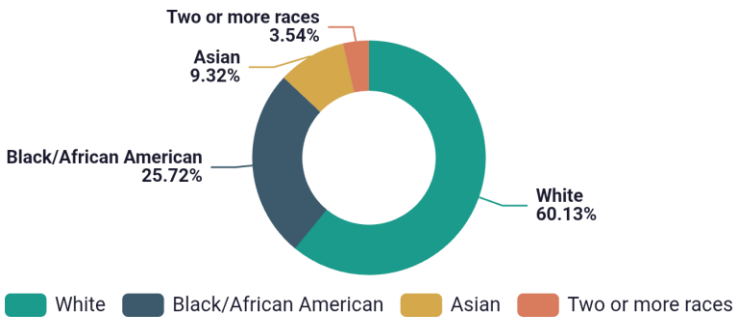
Gender Distribution



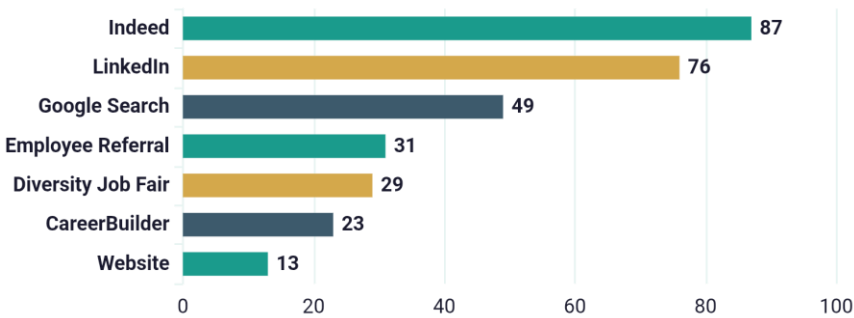
Marital Status Distribution



Race Distribution



Recruitment Source Performance



Top Termination Reasons

20

Another Position

14

Unhappy

11

More Money

9

Career Change

8

Hours

Employee Details Dashboard

Drill-down analytical capabilities for individual employee review

Dashboard Functionality

This dashboard serves as an **analytical lookup page for deeper HR review**, providing comprehensive individual employee information with interactive filtering capabilities.



Department Filter

Filter by department to view employees within specific organizational units



Employee Search

Search and select individual employees by name for detailed review



Comprehensive Details

View complete employee profile including position, salary, manager, and performance

Use Cases

Performance Review

Quickly access employee performance scores during evaluation meetings

Exit Interview Analysis

Review terminated employee details to identify patterns and trends

Employee Information Display

Employee Name

Adinolfi, Wilson K



Position

Production Technician I

Department

Production

Gender

Male (M)

Marital Status

Single

Salary

\$63,000

Performance Score

Exceeds

Manager Name

Michael Albert

Date of Hire

7/5/2011

Date of Birth

07/10/1983

08 Business Insights & Key Findings

Critical analytical discoveries from comprehensive workforce data analysis

Overall Attrition Rate

33%

One-third of the workforce has turned over, indicating significant retention challenges that require immediate strategic intervention. This rate is above industry benchmarks for most sectors.

Department Attrition Leaders

Production 40%
Highest risk department

Software Engineering 36%
Second highest risk

Recruitment Source Analysis

Google Search and Indeed showed the highest employee exits, suggesting potential issues with candidate-job fit or expectations management from these channels.

30
Google Search

21
Indeed

Performance vs Attrition

Attrition was NOT strongly performance-driven. Most exits came from the "Fully Meets Expectations" category, indicating that performance management alone won't solve retention issues.

 Key Insight: Focus on engagement and satisfaction, not just performance

Primary Termination Drivers

Unhappiness
Workplace satisfaction issues

Career Change
Limited growth opportunities

Compensation
Salary competitiveness

Work Hours
Work-life balance concerns

 **Critical Finding:** Demographic factors do NOT strongly drive attrition. Exits are more related to job satisfaction, growth opportunities, and workplace conditions rather than age, gender, or race.

09

Strategic Business Recommendations

Actionable HR strategies based on data-driven insights

1

Targeted Retention Plans

Implement department-specific retention strategies for Production (40% attrition) and Software Engineering (36% attrition). Conduct exit interviews to understand department-specific pain points.

✓ Priority: High

2

Engagement Surveys

Conduct department-level engagement surveys to identify specific factors contributing to unhappiness and dissatisfaction. Use pulse surveys for continuous feedback.

✓ Priority: High

3

Recruitment Channel Review

Reevaluate hiring effectiveness from Google Search and Indeed. Analyze candidate quality, job fit, and long-term retention rates by source to optimize recruitment spend.

✓ Priority: Medium

4

Enhanced Screening Process

Strengthen screening and interview processes to improve candidate-job fit. Implement behavioral assessments and culture fit evaluations to reduce early turnover.

✓ Priority: Medium

5

Performance Tracking

Track long-term performance of new hires from high-exit recruitment sources. Monitor 6-month and 12-month retention rates to identify problematic channels.

✓ Priority: Medium

6

Work-Life Balance

Improve work-life balance initiatives and address concerns related to work hours and workload management. Consider flexible scheduling and remote work options.

✓ Priority: High

Implementation Roadmap



Immediate (0-30 days)

Launch engagement surveys

Short-term (1-3 months)

Retention plans for high-risk depts

Medium-term (3-6 months)

Recruitment channel optimization

Long-term (6+ months)

Culture and policy transformation

Conclusion & Project Impact

Transforming HR data into strategic workforce intelligence



Project Success

This Power BI HR Analytics project successfully transformed raw workforce data into strategic intelligence, enabling data-driven decision-making for talent retention and organizational health management.



Business Value

The interactive dashboards provide HR leadership with real-time visibility into attrition patterns, demographic trends, and performance correlations—supporting proactive workforce management and sustainable business growth.

★ Key Achievements



Real-time Visibility

Live workforce metrics



Targeted Insights

Department-level analysis



Actionable Intelligence

Strategic recommendations



Scalable Solution

Cloud-based deployment

Thank you for your attention